Public Comment on the NMFS Advance Notice of Proposed Rulemaking

North Atlantic Right Whale Vessel Strike Reduction Rule (50 CFR 224.105)

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Executive Summary

The empirical evidence and population-modeling results analyzed support **retaining and expanding the existing mandatory speed rule** rather than replacing it with a primarily technology-based or voluntary dynamic regime. Available detection technologies could augment but not replace the successful speed rule in a **supportive regulatory architecture** that:

1. Retains the mandatory 10-knot speed limit in Seasonal Management Areas (SMAs)
2. Extends mandatory speed restrictions to vessels ≥ 35 ft (10.7 m) per the 2022 Preferred Alternative
3. Layers a mandatory Dynamic Speed Zone (DSZ) program atop static SMAs, triggered by TRL-9 offboard detection (PAM, aerial surveys) and delivered via real-time electronic notification
4. Codifies a calibrated safety-deviation provision (50 CFR 224.105(c)) with class-specific parameters and standardized reporting
5. Funds the unstudied vessel-avoidance-maneuver research gap as a near-term agency priority
6. Incentivizes onboard detection technology adoption as a complement, not a replacement

The North Atlantic right whale population numbered an estimated **384 individuals at the start of 2024** (NARW Consortium, +10/-9 CI), a modest 7% recovery from the 2020 trough of 358 but well below biologically safe levels. Population viability modeling calibrated to the observed 2011–2024 trajectory (mean absolute error 2.0%) indicates the species has no margin for regulatory experimentation that depends on unproven detection-and-avoidance assumptions.

1. Background and Method

1.1 The 2008 Speed Rule

Under 50 CFR 224.105, most vessels ≥ 65 ft (19.8 m) piloting through designated Seasonal Management Areas (SMAs) along the U.S. East Coast must travel at 10 knots or less during specified seasonal periods. National Marine Fisheries Service (NMFS) implemented the rule in 2008 (73 FR 60173) and removed the five-year sunset provision in 2013 (78 FR 73726).

1.2 The 2008 Voluntary DMA / 2020 Slow Zone Program

Concurrent with the speed rule, NMFS implemented a voluntary Dynamic Management Area (DMA) program, rebranded as Slow Zones in 2020, requesting voluntary 10-knot transits when sightings or acoustic detections indicate NARW aggregations outside SMA boundaries.

1.3 Analytical Framework

The analysis that follows uses a Monte Carlo population viability model (`narw_model_v2a.py`) calibrated to:

- **Population estimates 2011–2024** (NARW Consortium / NOAA Linden 2024 model)
- **MITRE/NOAA TRL Report MTR250363 (Nov 2025)** for technology parameters
- **Vanderlaan & Taggart (2007), Conn & Silber (2013), Lisi et al. (2025)** for speed–size–lethality
- **Blondin et al. (2025)** for vessel-class encounter risk apportionment

The model reproduces the observed 2011–2024 trajectory with **mean absolute error 2.0% and maximum error 6.6%** across three regimes before, during, and after the North American Right Whale Unusual Mortality Event (UME) between 2017-2020. The three regimes are labeled pre-UME 2011–2016; UME 2017–2020; recovery 2021–2024.

2. Findings

2.1 Speed-Rule Efficacy

Period	Documented vessel-strike deaths/serious injuries
1998–2007 (pre-rule)	12
2008–2017 (post-rule)	8

Inside-SMA effectiveness has been demonstrated by Laist et al. (2014) and van der Hoop et al. (2015) — vessel-strike mortalities effectively dropped to near-zero **inside** active SMAs after 2008. Outside-SMA mortalities continued at roughly the pre-rule rate, demonstrating that protection is bounded by SMA geographic and temporal extent rather than whale presence.

The variables of speed and size of the vessel compose the following conditional probabilities of lethality given that the vessel strikes a North American right whale.

Speed	Size 20–108 m P(lethal | strike),	Size ≥ 108 m P(lethal | strike)
10 kts	**27%**	86%

Speed	Size 20–108 m P(lethal | strike),	Size ≥ 108 m P(lethal | strike)
14 kts	73%	91%
16 kts	**88%**	93%

For mid-class commercial vessels, the 10-knot limit converts a strike from ~88% lethal to ~27% lethal — **a 3.3× reduction per encounter**. Large ocean-going ships (≥ 108 m) operate in a mass-dominated regime in which speed reductions provide only marginal benefit (Lisi et al. 2025; Kelley et al. 2021); for these vessels, **routing changes are the only means to protect the population of the whales**.

2.2 Compliance Performance

Regime	Approximate Compliance
Mandatory SMA rule (2008–2009)	~50%
Mandatory SMA rule (2018–2019)	**~81%**
Voluntary DMA / Slow Zones	**~10–30%**
Largest commercial vessels at four SE port entrances	**< 25%**

The 50–60 percentage-point gap between mandatory and voluntary compliance gives the strongest empirical evidence **against** replacing the static SMA rule with a primarily voluntary dynamic regime. Mandatory SMA compliance has steadily improved (50% → 81%) over the rule's first decade, indicating cultural normalization. That compliance **is not at a hard ceiling**.

2.3 Detection Technology Maturity (MITRE TRL Report)

Layer	Technology	TRL	Avg detection probability	Key limitation
Onboard	Thermal/IR	6	35%	Surface-only; degrades in fog/rain/seas
Onboard	Visual Camera	4	8%	Daytime only; AI-training gap
Onboard	Radar	3	5%	High false-alarm rate
Onboard	Active Acoustic	3	9%	Range < 1 km; few NARW studies
Onboard	PAM	3	5%	Flow noise at speed; vocalization-dependent
Offboard	**PAM (regional)**	**9**	n/a (zone trigger)	**Mature**
Offboard	**Aerial survey**	**9**	n/a (zone trigger)	**Mature**

Combined onboard detection probability:

- Clear daylight: 71%
- Fog/rain: 30%
- Night/poor visibility: 24%

After applying the 1.5 km range-sufficiency filter (the minimum warning distance needed to maneuver at 16 kts), **effective detection drops to ~35%**. With 50% assumed avoidance success, **effective per-encounter protection is ~18%** — substantially below the 73% lethality reduction the speed rule already delivers for mid-class vessels.

2.4 The Avoidance-Maneuver Research Gap

The MITRE report's most consequential finding is that **vessel maneuvers in response to whale detections have never been studied or documented**. The model therefore caps assumed avoidance-success probability at `MAX_AVOIDANCE_SUCCESS = 50%` ; values above 50% are not empirically defensible as field studies do not yet exist. Across the entire defensible 10%–50% sensitivity range, our simulation shows onboard technology underperforms the speed rule by **12–14% on long-run population**:

The model emits the conditional probabilities of avoiding a whale once detected and the projected final population after 50 years and the difference compared to keeping the current speed limit.

P(avoid|detect)	Mean final pop	Δ vs Speed Limit
10%	546	−88
20%	548	−86
30%	552	−82
40%	555	−79
50% (cap)	**560**	**−74**

The policy gap varies by only ~14 whales across this entire band, indicating the speed rule's superiority is **robust to uncertainty in the unstudied avoidance parameter**. This finding holds because the detect-then-maneuver chain has multiple failure modes (detection probability, range-sufficiency, decision time, hydrodynamic constraints, deep-draft / low-maneuverability physics) that speed reduction sidesteps simultaneously — an advantage that no level of avoidance success in the defensible range can recover.

2.5 Parametric Uncertainties Lead to Higher Final Population Means.

- The model optimistically assumes no UMEs that would disrupt the recovery rates.
- The model partitions the historical data into a shorter UME ending 2020 and the recovery period commencing in 2021. NOAA's current endpoint for the UME is set at 2026, therefore the model may produce higher estimates of North Atlantic right whales abundance.

2.6 50-Year Population Projections

With recovery-phase parameters (the regime that has held since 2021), calibrated to reproduce 2011–2024 within 2.0% MAE:

Scenario	Median final pop	P(quasi-extinction < 50)
Status-quo Speed Rule	**635**	0%
Onboard Tech (no speed limit)	560	0%
Enhanced Regional (DMA + emerging tech)	583	0%

The status-quo speed rule produces the highest long-run population. Removing the speed rule and substituting onboard detection technology costs ~75 whales over 50 years — a 12% reduction. The Enhanced Regional scenario sits between the two: better than removing the rule, but worse than universal 10-knot enforcement.

3. Responses to Specific NMFS RFI Topics

3.1 Vessel-Size-Specific Risk Assessment

Position: The available evidence argues *against* loosening the rule for any vessel class.

- **≥ 108 m (≈ 350 ft) ocean-going ships** (Lisi et al. 2025) operate in a mass-dominated lethality regime where speed reductions provide diminishing returns. Strike probability is also elevated by deep draft (10–15 m) that prevents whale dive-avoidance and by stopping distances of 1–2 km that preclude evasive maneuvering, thus routing changes and DSZs are the principal avoidance options.
- **20–108 m (65–350 ft) commercial vessels** are in a speed-dominated lethality regime where the existing rule provides its largest absolute benefit (3.3× per-encounter lethality reduction).
- **< 20 m recreational and small-fishing vessels** are currently exempt but contribute documented strike mortality (Laist et al. 2001; Sharp et al. 2019), particularly to neonates and calves on the SE U.S. calving grounds (Montes et al. 2020). Strike risk for this class is severely under-quantified due to Automatic Identification System (AIS) coverage gaps that NMFS should address by extending AIS carriage requirements to ≥ 35 ft (Blondin et al. 2025 applied an explicit AIS under-representation correction factor for the 26–65 ft class of vessls).

Recommendation: Extend mandatory speed restrictions to most vessels 35 ft - 65 feet per the proposed rule 87 FR 46921 proposed in 2022 but withdrawn January 8, 2025.

3.2 Alternative Management Areas (Dynamic, Real-Time Notification)

Position: Dynamic management should be implemented as a **mandatory layer over** the existing static SMAs, not as a replacement.

Rationale: voluntary DMA / Slow Zone compliance has remained at 10–30% despite a decade of outreach, while mandatory SMA compliance has reached 81%. Replacing the static rule with a voluntary compliance would decrease **the chances of the species survival by in erasing the 50–60 percentage-point compliance.**

Specific recommendations:

1. Make Dynamic Speed Zones **mandatory** when triggered by confirmed visual or acoustic detections.
2. Use real-time electronic notification (Whale Alert, AIS broadcast, mariner email/SMS). Pilot data suggest a 10–25 percentage-point compliance uplift from electronic over passive notification.
3. Set notification latency targets ≤ 6 hours from confirmed detection to mariner alert.
4. Continue investing in TRL-9 offboard detection (PAM, aerial surveys) to drive zone triggering and improve detection responsiveness toward 0.7+.
5. Define minimum DSZ activation criteria by vessel size class to prevent unintended exclusion of smaller, AIS-invisible craft.

3.3 Voluntary DMAs / Slow Zones Effectiveness

The voluntary program serves a useful adjunctive role and should be retained where mandatory measures are not yet feasible. However, its demonstrated voluntary compliance (10–30%) is too low to serve as a primary management tool. AIS-based effectiveness studies (2018–2024) consistently show voluntary measures produce marginal speed-distribution shifts in active zones.

Recommendation: Maintain voluntary DMAs / Slow Zones as an adjunct, not a substitute for the mandatory speed rule. Track compliance via AIS metrics-of-record published quarterly.

3.4 Safety Deviation Provision (50 CFR 224.105(c))

Position: Provide greater clarity and operational flexibility while preserving the core protective intent.

Specific recommendations:

1. **Codify objective deviation criteria** to include explicit weather-based triggers (Beaufort sea state ≥ 5, sustained winds ≥ 25 kts, demonstrable swamping/capsizing/loss-of-steerage hazard) rather than the current open-ended "safe maneuvering speed" language.
2. **Standardize deviation reporting** through a 48-hour electronic filing requirement to enable longitudinal analysis of deviation rates and circumstances.
3. **Differentiate by vessel class.** If the rule is extended to 35–65 ft vessels, the deviation rate is expected to be substantially higher (~10% vs. ~3% for ≥ 65 ft) due to dynamic-stability and freeboard limitations of small craft. This class-specific calibration is essential to balance crew safety and whale protection.
4. **Define a maximum deviation speed.** During permissible deviations, vessels should not exceed an upper bound (we suggest 14 kts) to retain partial lethality reduction even during heavy-weather operation.
5. **Sunset and review.** Build in a 5-year review cycle to assess deviation-rate trends against documented strike incidents.

3.5 Technological Interventions

Position: No onboard technology, individually or combined, can replace the speed rule at present TRL maturity. Onboard technology should be incentivized as a **complement** that fills gaps the static rule cannot reach.

Highest-leverage near-term interventions (TRL 9, deployable now):

Intervention	TRL	Status
Whale Alert / Whalemap electronic notification scale-up	9	Operational

Intervention	TRL	Status
Mandatory AIS for 35–65 ft vessels	9	Regulatory
Expanded PAM + aerial survey coverage	9	Operational
Mandatory Dynamic Speed Zones triggered by 1–3 above	9	Pilot-ready

Medium-term interventions (TRL 5–7 with investment):

Intervention	TRL	Status
Thermal/IR onboard cameras as complement to SMA rule	6	Demonstrating
Real-time bow-mounted detection alerting	5–6	Demonstrating
Smart routing (AIS + Whalemap + weather fusion)	6–7	Pilots underway

Long-term R&D priorities:

Research priority	Why
Avoidance-maneuver field studies	**Closes the single largest model uncertainty**
NARW-specific active-acoustic detection validation	Few NARW-specific studies exist
Flow-noise mitigation for onboard PAM at speed	Unlocks acoustic detection per vessel
Multi-sensor fusion algorithms with quantified false-alarm rates	Enables alert-fatigue management
Climate-prey-shift coupled distribution models	Forecasts future SMA boundaries

Economic incentives for adoption: We recommend NMFS coordinate with USCG and Customs to provide port-fee discounts, reduced inspection burdens, or insurance credits for vessels equipped with multi-sensor detection systems and operating in high-risk corridors, while maintaining the speed rule as the mandatory baseline.

4. Recommended Hybrid Regulatory Architecture

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         │   STATIC SMAs    │   │ DYNAMIC DSZs │   │   ONBOARD TECH   │
         │    (mandatory)   │   │  (mandatory) │   │  (complement)    │
         │   ≥ 35 ft, 10kt  │   │ PAM + Aerial │   │   Thermal/IR +   │
         │   85% compliance │   │ triggered;   │   │   Multi-sensor;  │
         │   (proven)       │   │ electronic   │   │   incentivized;  │
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                            │ Calibrated by class│
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Six-point program:

1. Retain the mandatory 10-knot SMA rule, expanded to vessels ≥ 35 ft per the 2022 Preferred Alternative.
2. Layer mandatory Dynamic Speed Zones triggered by TRL-9 offboard detection (PAM, aerial surveys), delivered via real-time electronic notification.
3. Codify a calibrated safety-deviation provision with class-specific parameters, weather triggers, and standardized 48-hour reporting.
4. Fund the avoidance-maneuver research gap as a near-term agency priority.
5. Incentivize onboard technology adoption as a complement, not a substitute.
6. Maintain voluntary DMAs / Slow Zones as an adjunct in unregulated areas while the mandatory framework expands.

5. Closing Statement

The North Atlantic right whale population reached an estimated **384 individuals at the start of 2024** — a 7% recovery from the 2020 trough of 358 but well below the levels needed for resilience against climate-driven prey shifts, episodic mortality events (such as the 2017 Gulf of St. Lawrence crisis), and stochastic environmental variability. Population viability modeling indicates that **only sustained recovery-phase parameters** (low entanglement mortality, elevated calving rate, low strike mortality) yield positive 50-year trajectories under any management scenario.

Weakening the speed rule's mandatory scope or compliance reach lead to risks of declining population to UME-era rates, losing approximately 25 whales per year — an unrecoverable trajectory for a population of this size.

The species has no margin for regulatory experimentation that depends on unproven detection-and-avoidance assumptions. We urge NMFS to **retain the speed rule's protective effect, expand its scope to more vessel classes, layer mandatory dynamic zones on top, and treat technology as a complement that fills gaps the static rule cannot reach**.

Respectfully submitted,

[Submitter name] [Affiliation] [Email] · [Phone] · [Mailing address]

References

1. Blondin, H., Garrison, L. P., Adams, J. D., Roberts, J. J., Good, C. P., Gahm, M. P., Lisi, N. E., & Patterson, E. M. (2025). Vessel strike encounter risk model informs mortality risk for endangered North Atlantic right whales along the United States east coast. *Scientific Reports* 15: 736. doi:10.1038/s41598-024-84886-z
2. Conn, P. B. & Silber, G. K. (2013). Vessel speed restrictions reduce risk of striking the endangered North Atlantic right whale. *Ecosphere* 4(4):43. doi:10.1890/ES13-00004.1
3. Crum, N., Gowan, T., Krzystan, A., & Martin, J. (2019). Quantifying risk of vessel strike on right whales using a discrete-time Cormack- Jolly-Seber model. *Ecosphere* 10(7): e02780.
4. Garrison, L. P., Adams, J., Patterson, E. M., & Good, C. P. (2022). Assessing the risk of vessel strike on North Atlantic right whales. *Endangered Species Research* 47: 117–135.
5. Kelley, D. E., Vlasic, J. P., & Brillant, S. W. (2021). Assessing the lethality of ship strikes on whales using simple biophysical models. *Marine Mammal Science* 37(1): 251–267.
6. Laist, D. W., Knowlton, A. R., Mead, J. G., Collet, A. S., & Podesté, M. (2001). Collisions between ships and whales. *Marine Mammal Science* 17(1): 35–75.
7. Laist, D. W., Knowlton, A. R., & Pendleton, D. (2014). Effectiveness of mandatory vessel speed limits for protecting North Atlantic right whales. *Endangered Species Research* 23: 133–147.
8. Linden, D. W. (2024). Population size estimation of North Atlantic right whales from 1990–2023. NOAA Technical Memorandum.
9. Lisi, N. E., Good, C. P., Garrison, L. P., Gahm, M., Patterson, E. M., & Blondin, H. (2025). The effects of vessel speed and size on the lethality of strikes of large whales in U.S. waters. *Frontiers in Marine Science* 11: 1467387. doi:10.3389/fmars.2024.1467387
10. Kirsch, C. C., Weber, T., & Adams, M. MITRE Corporation (2025). Technology Readiness Level Report (TRL) for North Atlantic Right Whale Detection and Vessel Strike Risk Reduction. MTR250363. November 2025
11. Montes, N. L., Swett, R., & Gowan, T. A. (2020). Risk of encounters between North Atlantic right whales and recreational vessel traffic in the southeastern United States. *Ecology and Society* 25(4):12. doi:10.5751/ES-11923-250412
12. NMFS. (2020). North Atlantic Right Whale Vessel Speed Rule Assessment. NOAA Office of Protected Resources, June 2020.
13. NMFS. (2022). Draft Environmental Assessment for Amendments to the North Atlantic Right Whale Vessel Strike Reduction Rule. NOAA Office of Protected Resources, July 2022.
14. Pace, R. M. III, Corkeron, P. J., & Kraus, S. D. (2017). State–space mark–recapture estimates reveal a recent decline in abundance of North Atlantic right whales. *Ecology and Evolution* 7(21): 8730–8741.
15. Pace, R. M. III, Williams, R., Kraus, S. D., Knowlton, A. R., & Pettis, H. M. (2021). Cryptic mortality of North Atlantic right whales. *Conservation Science and Practice* 3(2): e346.
16. Sharp, S. M., McLellan, W. A., Rotstein, D., et al. (2019). Gross and histopathologic diagnoses from North Atlantic right whale *Eubalaena glacialis* mortalities between 2003 and 2018. *Diseases of Aquatic Organisms* 135(1): 1–31.
17. Silber, G. K., Slutsky, J., & Bettridge, S. (2010). Hydrodynamics of a ship/whale collision. *Journal of Experimental Marine Biology and Ecology* 391(1–2): 10–19.
18. Silber, G. K., Vanderlaan, A. S. M., Tejedor Arceredillo, A., Johnson, L., Taggart, C. T., Brown, M. W., Bettridge, S., & Sagarminaga, R. (2012). The role of the International Maritime Organization in reducing vessel threat to whales: process, options, action and effectiveness. *Marine Policy* 36(6): 1221–1233.
19. Vanderlaan, A. S. M. & Taggart, C. T. (2007). Vessel collisions with whales: the probability of lethal injury based on vessel speed. *Marine Mammal Science* 23(1): 144–156. doi:10.1111/j.1748-7692.2006.00098.x
20. van der Hoop, J. M., Vanderlaan, A. S. M., Cole, T. V. N., Henry, A. G., Hall, L., Mase-Guthrie, B., Wimmer, T., & Moore, M. J. (2015). Vessel strikes to large whales before and after the 2008 Ship Strike Rule. *Conservation Letters* 8(1): 24–32.

Appendix A — Population Viability Modeling Methodology

The quantitative claims in this comment are supported by a Monte Carlo population viability model implemented in `narw_model_v2a.py`. Key methodological elements:

- **Demographic structure:** Logistic births capped by carrying capacity K=1000; binomial natural and entanglement mortality; Poisson encounter and strike processes.
- **Strike lethality:** Three-regime function (`strike_lethality`) following Lisi et al. (2025) with vessel-size-class branches: ≥ 108 m (mass-dominated), 20–108 m (Vanderlaan & Taggart 2007 logistic), < 20 m (right-shifted recreational).
- **Strike probability:** Linear-in-speed term with multiplicative draft and maneuverability modifiers (`strike_probability_given_encounter`) parameterized from Blondin et al. (2025).
- **Hindcast calibration:** Phase-varying parameters (pre-UME 2011–2016, UME 2017–2020, recovery 2021–2024) reproduce the observed NARW Consortium / NOAA series within mean absolute error 2.0% and maximum error 6.6%.
- **Forward projections:** 50-year horizons starting from 2024 N_0=384, with 2,000 Monte Carlo runs per scenario.
- **Sensitivity analysis:** Avoidance-success rate swept across the empirically defensible range [10%, 20%, 30%, 40%, 50%]. Values above 50% (`MAX_AVOIDANCE_SUCCESS`) are not reported because the unstudied status of vessel-avoidance maneuvers (MITRE 2025) does not support them. The cap is enforced via dataclass `__post_init__` so any caller passing values > 50% receives the capped value, preventing accidentally indefensible runs.

Source code, calibrated parameters, and full reference bibliography are available at:
https://github.com/SHodgeComputer/Narw